

Advanced Data Communication: Principles and Applications of Satellite

Acquah Baffoe Kofi

Pentecost University

PBIT402: Advanced Data Communication

Dr. Harry Tieku

May 1, 2026

1 CONTENTS

Abstract	4
1 Introduction	5
1.1 Evolution of Satellite Communication	5
1.1.1 Role of Satellite in data infrastructure	10
2 Technical Principles: Analysis of orbital mechanics (LEO, MEO, GEO)	13
2.1.1 Low Earth Orbit (LEO)	13
2.1.2 Medium Earth Orbit	13
2.1.3 Geostationary Orbit	14
2.1.4 Frequency Management Spectrum	14
2.1.5 Use of C-band in satellite communication	15
2.1.6 Ku band	16
2.1.7 ka band	17
2.1.8 Signal modulation	18
3 Satellite Zooming & Precision	21
3.1.1 Satellite zooming	21
3.1.2 Technical exploration of High-Throughput Satellites (HTS).	21
3.1.3 frequency Reuse	24
3.1.4 Spot Beam	24
4 Comparative Analysis	25
4.1.1 Advantage of satellite	25
4.1.2 Disadvantages of satellites	25
4.1.3 Disadvantages of Terrestrial Data Communication	26
4.1.4 Challenges in Satellite Data Communication	27
5 Future Trends	28

ABSTRACT

Satellite communication is an essential part of the worldwide data network. In this paper, the topic of satellite data communication systems is thoroughly investigated from the historical perspective of how satellite technology has evolved into its current function of interconnection. Technical grounds include the study of Low Earth Orbit, Medium Earth Orbit, and Geostationary orbit principles; C, Ku, and Ka-band spectra control; and modulation techniques used in the process.

This paper examines the concept of Satellite Zooming or High Throughput Spot Beam technology, which explains frequency reuse in spot beams distribution. The author gives comparisons of satellite versus ground data communication systems in terms of their strengths and weaknesses. Among the four major problems, there are atmospheric attenuation, latency, orbital debris, and cybersecurity. All of these obstacles make satellite communication rather challenging.

As for the future trends, it is expected that new satellite technologies would facilitate the integration of AI-powered 6G network. Conclusion – satellite systems will keep functioning as one of the key components of global data communication.

1 INTRODUCTION

This part of research essay will explore the term satellite, satellite communication, the evolution of satellite communication and the role of satellite in modern data infrastructure. A satellite is an object in space that revolves or circles around a bigger object. There are two kinds of satellites: natural, such as the moon orbiting the Earth or artificial such as the International Space Station orbiting the Earth. There are dozens upon dozens of natural satellites in the solar system, with almost every planet having at least one moon. Saturn, for example, has at least 53 natural satellites, and between 2004 and 2017, it also had an artificial one — the Cassini spacecraft, which explored the ringed planet and its moons. (Howell, 2022)

satellite communication, in telecommunications, is the use of artificial satellites to provide communication links between various points on Earth. Satellite communications play a vital role in the global telecommunications system. Approximately 2,000 artificial satellites orbiting Earth relay analog and digital signals carrying voice, video, and data to and from one or many locations worldwide.

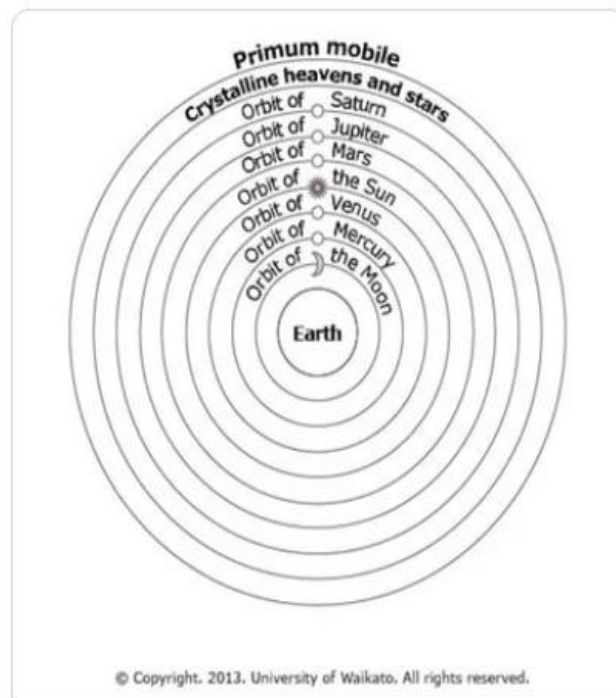
Satellite communication has two main components: the ground segment, which consists of fixed or mobile transmission, reception, and ancillary equipment, and the space segment, which primarily is the satellite itself. A satellite link involves the transmission or up-linking of a signal from an Earth station to a satellite. The satellite receives and amplifies the signal and with re-transmission it is sent back to Earth, where it is been received and re-amplified by Earth stations and terminals. Satellite receivers on the ground include direct-to-home (DTH) satellite equipment, mobile reception equipment in aircraft, satellite telephones, and handheld devices. (National Aeronautics and Space Administration [NASA], 2023).

1.1 EVOLUTION OF SATELLITE COMMUNICATION

Geocentric model

Ancient Greek astronomers believed that the Earth is the center of the Universe and that all bodies orbit the Earth. This idea was accepted in ancient times since the sky looked as if it revolved around as.

The first astronomer to bring this idea in was Anaximander. Other astronomers, such as the famous Aristotle and Claudius Ptolemy, also developed more complex geocentric

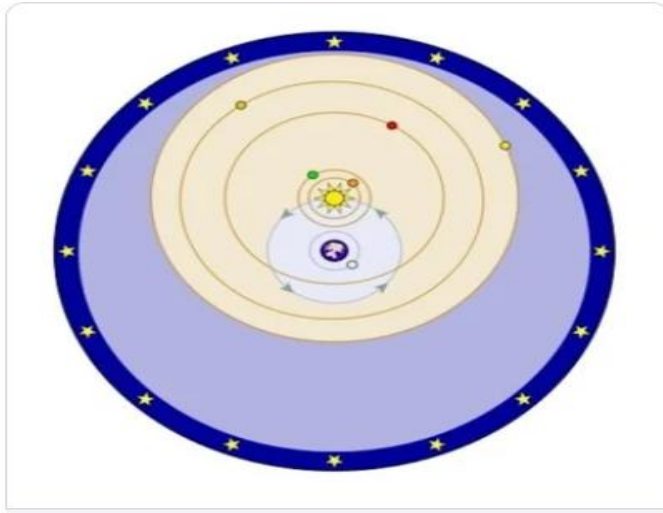


models

An astronomer called Ibn al-Satir, a Damascene astronomer, accepts the geocentric model but made observations and experimentation that challenge the Ptolemaic model. He did that through calculations. And his idea of the moon arrangement is similar to that of another scientist, who was called Copernicus after his death.

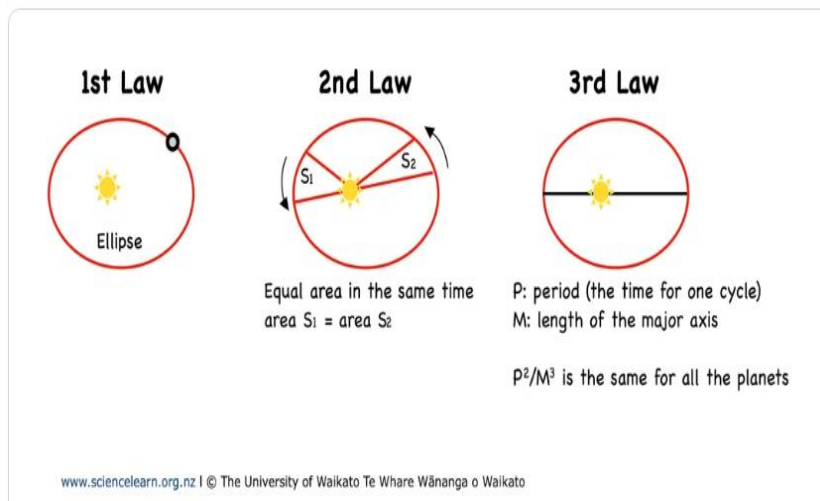
Polish astronomer Copernicus was a churchman from Poland known by his Latin name who assumed that the Sun is stationary in the Centre of the Universe and the Earth and other planets revolve around it. The church oppressed this controversial idea, but it brought change in astronomy (Westman, R. S., 1975)

Danish astronomer Tycho Brahe built instruments that made him to view stellar and planetary positions. His records of the motion of Mars are later used by Kepler



Italian astronomer Galileo Galilee's improvements to the telescope allow him to observe Venus's phases, the largest satellites of Jupiter, a supernova and sunspots. His discoveries prove the Copernican heliocentric system. The Roman Inquisition finds him guilty of heresy.

German astronomer Johannes Kepler (1571–1630) uses the term 'satellite' to describe the moons orbiting Jupiter. He develops the three laws of planetary motion, and his accurate astronomical tables provide evidence for the Copernican heliocentric model.



1687 – Newton and gravity

Sir Isaac Newton publishes *Principia* in which he states the three laws of motion and describes universal gravity. This lays the foundation for our understanding of

The first fictional depictions of satellites being launched into orbit are published in Edward Everett Hale's short story *The Brick Moon* (1869) and Jules Verne's *The Begum's Fortune* (1879).

1903 – First orbit of Earth calculated

Russian Konstantin Tsiolkovsky (1857–1935) proposes using rocketry to launch spacecraft. He calculates the orbital speed required for a minimal orbit around the Earth at 8 km/s and that a multi-stage rocket fueled by liquid propellants could be used to achieve this.

1928 – First description of a space station

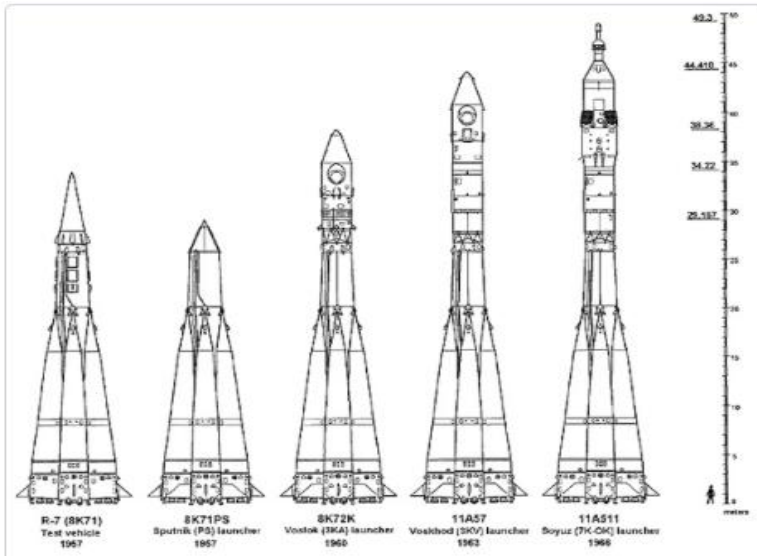
Slovenian Herman Potočnik (1892–1929) describes geostationary satellites and communication between them and Earth using radio. He also describes a space station in detail and the use of orbiting spacecraft for detailed observation of the Earth and scientific experiments.

1945 – Satellite communications system proposed

British science fiction writer and inventor Arthur C Clarke (1917–2008) publishes an article that shows how geostationary satellites could be used for worldwide radio and television broadcasts and communication.

1957 – First artificial satellite launch

The Soviet Union launches Sputnik, the first artificial satellite to orbit around the Earth. It has a mass of 83.6 kg and travels in an elliptical orbit at a height above the Earth between 939 km and 215 km. It travels at 29,000 km/h and takes 96.2 minutes for each orbit.



1960 – First successful weather satellite

NASA launches Tiros-1, the first weather satellite considered a success. It transmits infrared images of Earth's cloud cover and is able to detect and chart hurricanes. This begins the Trios program, which is followed by the Nimbus program me of weather satellites. (University of Waikato, 2022)

1961 – First person to orbit Earth

Russian cosmonaut Yuri Gagarin (1934–1968) was the first person to orbit the Earth in his Vostok spacecraft.

1962 – First communications satellitethe

The first lived transatlantic telecast was broadcast using Telstar-1, a low-orbit satellite. Later, an international global satellite consortium (Intelsat) was formed, which manages a constellation of communications satellites providing international broadcast services. (University of Waikato, 2022)

1.1.1 ROLE OF SATELLITE IN DATA INFRASTRUCTURE

Satellite has basically changed global connectivity by expanding data communication infrastructure to cities and villages at places like Africa including Ghana, Togo and Nigeria etc. Where connectivity would have been difficult. (TelecomWorld101.com, 2024)

Satellites have made it easier to manage disasters due to their capacity to establish connectivity on a global scale, it has also been reliable in communication situations where existing communications systems can be disrupted. Satellites enhance easy coordination between emergency response and the spreading of important information to those affected by the disaster.

Internet Facilitation Around the Globe

Satellite has also made it easy to connect to others at farther ranges. For example, connecting with a brother abroad or my grandmother in a village in some part of Ghana can be possible as a result of satellite communication.

Internet access is made even at distant locations through satellites, therefore making communication or information sharing easier

Broadband Connectivity

Communication satellites offer broadband connectivity in regions that do not have terrestrial connectivity, which has resulted in enhancing the spread of global internet access.

Data Transmission

Satellite communication has made it possible for people to chat live and track whenever they are improving experience and security

Technical Principles

A satellite is an independent communications system with the capacity to work in orbit above the Earth. A satellite cannot depend on external support once it is deployed into orbit (Elbert, 2004). It must generate its own power through solar panels, regulate its own systems through onboard computers, and sustain uninterrupted operation potentially for decades (Roddy, 2006; Freeman,

2007). The transponder is the operational component of every communication satellite. It receives signals from Earth ground stations and retransmits those signals back with the help of a transponder, an integrated unit that merges receiver and transmitter in one device. (Pratt, Bostian & Allnutt, 2003). Satellites are one of the most demanding engineering technologies. As Roddy (2006) noted, a satellite must endure extraordinary physical and environmental stresses from the moment of launch through its entire operational lifespan, which can extend up to 20 years

The first challenge a satellite faces before it even reaches orbit. According to Pratt et al. (2003), during deployment, a satellite must withstand the intense mechanical shock of being accelerated to an orbital velocity of approximately 28,100 km per hour (17,500 miles per hour). This level of acceleration places tremendous structural stress on every component of the satellite, requiring its physical construction to be uniquely robust before it even begins its operational life (Freeman, 2007).

When the satellite is set to orbit, the problems do not reduce. Elbert (2004) says the space environment is hostile, exposing satellites to intense radiation and temperature changes that ground-based systems would never face. These conditions are continuous throughout the satellite's orbiting lifespan. Hence, Haykin (2001) says that every material and component used in satellite system construction must be selected with long-term environmental resistance in mind.



Intelsat VI Intelsat VI, a communications satellite, after being repaired, 1992.

The main components of a satellite consist of the communications system, which includes the antennas and transponders that receive and re-transmit signals, the power system, which includes the solar panels that provide power, and the propulsion system, which includes the rockets that propel the satellite. A satellite needs its own propulsion system to get itself to the right orbital location and to make occasional corrections to that position. A satellite in geostationary orbit can deviate up to a degree every year from north to south or east to west of its location because of the gravitational pull of the Moon and Sun. A satellite has thrusters that are fired occasionally to adjust in its position. The maintenance of a satellite's orbital position is called "station keeping," and the corrections made by using the satellite's thruster are called "attitude control." A satellite's life span is determined by the amount of fuel it has to power these thrusters. Once the fuel runs out, the satellite eventually drifts into space and out of operation, becoming space debris.

A satellite in orbit has to operate continuously over its entire life span. It needs internal power to be able to operate its electronic systems and communications payload. The main source of power is sunlight, which is harnessed by the satellite's solar panels. A satellite also has batteries on board to provide power when the Sun is blocked by Earth. The batteries are recharged by the excess current generated by the solar panels when there is sunlight.

Satellites operate in extreme temperatures where the temperature may be very hot or very cold. And they are also faced with radiation from the sun and other activities. To prevent satellites from getting spoiled, they are protected from radiation and heat with elements such as aluminum shields to make them resist radiation. Satellite also contain in built cooling system to prevent spoiling from extreme temperatures and also a heating system to heat up the system in times of coldness. The thermal system ensures satellite protection.

The satellite's tracking, telemetry, and control system is a two-way communication link between the satellite and the ground-based tracking, telemetry, and control system. This allows a ground station to view the position of a satellite and to control the satellite's movement, thermal, and other systems in the satellite. It can also monitor a satellite's temperature, electrical voltages, and other important components of the satellite.

Communication satellites are different in size and mass. Micro-satellites weigh less while the largest satellites weigh heavy. This range shows the various purposes satellites serve, from small experimental missions to large-scale commercial broadcasting (Britannica, 2025).

Satellites have three different orbits it moves in low Earth orbit, medium Earth orbit, and geostationary. Low Earth Orbit satellites are positioned at an altitude between 160 km and 1,600. Medium Earth orbit satellites operate from 10,000 to 20,000 km. Satellites do not operate between Low Earth Orbit and Medium Earth orbit because of the state of the environment that is harmful to electronic components in that area this said to be caused by the Van Allen radiation belt.) Geostationary Earth Orbit satellites are positioned 35,786 km above Earth, where they complete one orbit in 24 hours. Few geostationary Earth orbits.

A signal that is bounced off a GEO satellite takes approximately 0.22 seconds to travel at the speed of light from Earth to the satellite and back. This delay poses some problems for applications such as voice services and mobile telephony. Therefore, most mobile and voice services usually use LEO or MEO satellites to avoid the signal delays resulting from the inherent latency in GEO satellites. GEO satellites are usually used for broadcasting and data applications because of the larger area on the ground that they can cover.

Launching a satellite into space requires a very powerful multistage rocket to propel it into the right orbit. Satellite launch providers use proprietary rockets to launch satellites from sites such as the Kennedy Space Center at Cape Canaveral, Florida, the Baikonur Cosmodrome in Kazakhstan, Kourou in French Guiana, Vandenberg Air Force Base in California, Yichang in China, and Tangshan Island in Japan.

2 TECHNICAL PRINCIPLES: ANALYSIS OF ORBITAL MECHANICS (LEO, MEO, GEO)

2.1.1 LOW EARTH ORBIT (LEO)

Low Earth Orbit is the closest orbit to the Earth's surface. Low Earth Orbit satellites reduce launch costs, have minimal signal lag, and have high-resolution imaging capability. They are also exposed to atmospheric drag and require significant power to sustain their orbits, reducing their lifespan to around 7 to 10 years. Low Earth Orbit satellites are best for telecommunications due to tracking difficulties, but networks of linked satellites can overcome this challenge. (Britannica, 2025).

2.1.2 MEDIUM EARTH ORBIT

Medium Earth Orbit is the orbital region between Low Earth Orbit and Geostationary Earth Orbit, known as home to navigation and GNSS satellites at approximately 20,000 km above Earth. Medium Earth Orbit satellite constellations are increasingly being used to send high-speed, low-latency connectivity that mirrors the performance of fibre optic cables. This makes Medium Earth Orbit is used for serving remote and underserved environments from ships and aircraft to offshore platforms and disaster relief zones. Ground-based infrastructure is impossible to deploy (SES, 2025).

2.1.3 GEOSTATIONARY ORBIT

Geostationary orbit is a circular orbit 35,785 km above Earth. A satellite in this orbit sees Earth as stationary in the sky. This particular orbit is used for meteorological and communications satellites. The geostationary orbit rotates at the same speed as the Earth. The concept for such an orbit was proposed in 1945 by British author and scientist Arthur C. Clarke in an article entitled “Extra-Terrestrial Relays” for Wireless World. Clarke, A. C. (1945). The article described the current communications satellite system that relays radio and television signals throughout the world. The U.S. communications satellite Syncom 3, which was launched on Aug. 19, 1964, was the first object to be placed in geostationary orbit. NASA. (1964).

2.1.4 FREQUENCY MANAGEMENT SPECTRUM

Spectrum management is the process of controlling the frequencies to promote efficient use and gain benefits from it. The term radio spectrum refers to the frequency range from one Hertz (unit of measurement of frequency) to Terahertz that may be used for wireless communication. When there is an increase in demand for services such as mobile telephones, computers, laptops, and many other devices, that needed to be change the spectrum management. Because a lot of people need wireless broadband, it has resulted in technological innovation, such as 3G and 4G mobile services, and the rapid expansion of wireless internet services to meet the demands of consumers.

2.1.5 C-BAND IN SATELLITE COMMUNICATION

The C band was the first frequency band that was provided for commercial telecommunications by means of satellite. Before its inception, the same frequency was used by terrestrial stations on the grounds with cables and towers to ensure the bands reach their destination.

The C-band communication satellites have 24 radio transponders which works as the links between the ground station and the satellite by receiving frequencies from the ground station, amplifying the frequencies and sending them back to the earth ground stations. Transponders on the same polarization are always 40 MHz apart. Out of the forty hertz, each transponder uses about 36 MHz of it. The 4Hz, which is between transponders, acts as a guard band for the imperfections in the microwave electronics to prevent interference between near transponders. This ensures a clear signal and which are of no overlapping.

One use of the C band is to ensure open satellite communication. For example, telephone and data networking. This use is opposite to that of direct-broadcast satellite, which is a completely closed system used to deliver subscription programming to small satellite dishes that are connected with proprietary receiving equipment.

The satellite communications portion of the C band has to deal with television satellite reception systems which is known as "big dish" systems, because antennas which are larger can not perform best the C band. The C band is usually received by larger dishes. For satellite communications, the frequencies of the C band is effective under bad weather conditions as compared with the Ku band microwave frequencies used by other communication satellites. The adverse conditions that have effect on transmission is collectively know as rain fade which are etc.

The C band is used in the health sector ,for example medicals cause by moist in the air and industrial heating applications and many unlicensed short-range frequency communication systems, such as phones, babies monitor, and keyless entry systems for vehicles. The C-band frequencies are used for Wi-Fi wireless connection in the 5 GHz spectrum. (Wikipedia, 2024)



2.1.6 KU BAND

The Ku band is the portion of the frequency spectrum that ranges from frequencies of about 12 to 18 gigahertz. It is mostly used in radar systems, and it ranges from 12 to 18 GHz according to the formal definition of radar frequency band nomenclature in IEEE Standard 521–2002. (IEEE Communications Society,2022).

Ku band is used for satellite communications. It is known for the downlink used by direct broadcast satellites to send satellite television, and for some applications, such as NASA's Tracking Data Relay Satellite, used for International Space Station (ISS) communications and SpaceX Starlink satellites. Ku band satellites are used to backhaul for satellite from remote locations back to a television network's studio for editing and broadcasting. The band is divided into different variations in different geographical regions.

NBC was the first television network to uplink its number of feeds through Ku band in 1983. Radar guns used by law enforcement to detect vehicles speeding make use of the Ku band

2.1.7 KA BAND IN SATELLITE COMMUNICATION

The Ka band is a portion of the frequency spectrum. It has not been defined by standards, its frequency ranges from 27 to 40 gigahertz in tables of the industry of electrical, electronic, and engineering(IEEE).

The International Telecommunication Union approves Ka band satellite networks in the 17.3 to 31 GHz frequency range, with Ka band satellite networks having uplinks in the 27.5 to 31 GHz and downlinks in the 17.7 to 21.2 GHz range. (Ka band - Wikipedia, n.d.; ITU, 1997). it is sometimes unusable when transmissions are in longer ranges

The Ka band, which ranges from 30 to 20 gigahertz, is used to transmit signals or data from ground stations to towers in the sky or to satellites in satellite communication. This band is employed in a security defense system. The military aviation used this band by integrating it into their airplanes' and fighter jet radar systems for targeting and by high resolution to capture clearly and for detection. (Ka band - Wikipedia, n.d.) The Kepler Mission used this frequency range to transmit the scientific data collected by the space telescope from the satellite to the ground station. This frequency is also used for remote sensing of clouds by radar, by both ground-based and satellite systems such as INCUS. (Investigation of Convective Updrafts). (Ka band - Wikipedia, n.d.). (eoPortal, 2023).

In satellite communications, the Ka band allows higher capacity transmission of data in communication. For example in ACTS Gigabit Satellite Network, and it is also used for high-throughput satellite Internet access in geostationary orbit (GEO) by the Inmarsat I-5 system, Pacific K-1 satellite, the Via-Sat 1, 2, and 3 satellites among others; in low Earth orbit (LEO) by the Space X Star link system and the Iridium Next satellite series; it is also used in medium Earth orbit (MEO) by the SES O3b system, the James Webb Space Telescope. (Wikipedia, n.d.)

The Ka band is more easily to be affected by rain attenuation than that of the Ku band, which in can also be affected easily than the C band. 5th generation mobile networks will also partially overlap with the Ka band (28, 38, and 60 GHz) (Wikipedia, n.d.).

2.1.8 SIGNAL MODULATION

Signal modulation is the process of modifying a carrier wave's properties — amplitude, frequency, or phase — to encode information for transmission. It enables the efficient transfer of data in wireless communication systems, broadcasting, and digital networks (IB-Lenhardt, n.d). There are different forms of modulation, each is designed to change a characteristic of the

Analog Modulation

The signal is encoded onto the carrier so that it conveys data over distances to the destination. Analogue carrier waves are continuous. For example, Amplitude Modulation and Frequency Modulation of the carrier wave vary. These were the earliest types of modulation, used to transmit audio signals in Amplitude Modulation and Frequency Modulation radio broadcasting (Wikipedia, n.d.).Wikipedia

Digital modulation

In digital modulation, a continuous carrier wave is converted into a binary digital signal. In Amplitude Shift Keying, the modulated wave shifts quickly between high amplitudes and low amplitudes. In Phase Shift Keying (PSK), amplitude and frequency remain constant, and the bit stream is represented by shifts in the phase of the signal (Britannica, 1998). Britannica

To transmit data and digitize information over a communications medium, an analog carrier wave can be converted to reflect the binary of the digital baseband signal. The portion of the carrier that can be changed are the amplitude, the frequency, and the phase.

Amplitude modulation

This modulation is used to send data such as sound, text, and images from one place to another using radio waves. (Britannica, 1998). Amplitude modulation is an old technique of broadcasting

radio programs. Commercial amplitude modulation stations operate in a frequency range of 10 kHz apart between 540 and 1,700 kHz (U.S. Code of Federal Regulations, n.d.). Radio waves in this frequency range are reflected to Earth's surface by the ionosphere and can be detected by receivers hundreds of kilometers away (Federal Communications Commission, 2015). It also transmits the video portion of television broadcasting

Frequency modulation

In frequency modulation, the amplitude of the carrier is kept constant, but its frequency is modified in relation to the differences in the signal being sent. This form of modulation was developed by the American electrical engineer Edwin H. Armstrong in the early 1930s in an effort to overcome interference and noise that affect the strength of radio signal reception. Frequency modulation cannot easily be interfered with by thunderstorms and by random electrical currents from machinery as much as Amplitude modulation. Noise signals can affect the amplitude of a radio wave but not its frequency. As a result of this, the frequency signal remains unchanged. (Britannica, 1998). Frequency modulation is best for sending stereo sound because frequency modulation carries two separate audio channels that allow sound to be heard from different directions (Diffen, n.d). The audio signals for television programs are sent using frequency modulation since it delivers clearer and higher quality sound compared to AM (Britannica, 1998). Frequency modulation is also used in microwave radio relay systems for long-distance telephone calls because its signal is more stable and less affected by interference over long distances (AudioDramaProduction, 2026). Commercial frequency modulation broadcasting stations are assigned higher frequencies than AM stations. The assigned frequencies, spaced 200 kHz apart, range from 88 to 108 MHz (Diffen, n.d).

Phase modulation

Phase modulation is the method of changing the amplitude and frequency of the wave. Phase modulation shifts the position of the wave to convey the information (Britannica, 1998).

The two processes are related because phase modulation cannot be changed without also varying the speed and strength of the wave. Also, the rate at which the phase of a carrier changes directly varies with the frequency of the audio signal.

Phase modulation is best used in situations where a clean, noise-free signal is needed at lower frequency ranges (Britannica, 1998).

Phase modulation can be applied during the amplification of a sound signal in broadcasting. Phase modulation is also used in Wi-Fi and satellite television systems.

Amplitude-shift keying

Amplitude shift key is when amplitude is the only chosen part of the carrier wave to change information signal. Amplitude shift key can be said as a digital version of analog amplitude modulation.

Frequency-shift keying

If frequency is the parameter chosen to be a function of the information signal, the modulation method is called frequency-shift keying (FSK).

Phase modulation

When the phase is the parameter changed by the information signal, the method is called phase-shift keying

Pulse modulation

Is a series of on-off pulses that act as the carrier wave converted to carry information. Pulse-Coded Modulation converts an information signal into a series of constant-amplitude pulses in coded patterns. Pulse-Coded Modulation reduces transmission loss and eliminates noise and interference problems since the receiving unit needs only to detect and find the pulse patterns. Pulse has equal amplitude height. In pulse duration modulation, information is in how long a pulse lasts. The order and repetition of the pulse carries intelligence.

3 SATELLITE ZOOMING & PRECISION

Satellite zooming refers to the ability to view and interact with satellite imagery at various zoom levels, allowing users to explore detailed geographical features and real-time data

3.1.1 SATELLITE ZOOMING

Satellite zooming refers to the ability to view and interact with satellite imagery at various zoom levels, allowing users to explore detailed geographical features and real-time data

3.1.2 TECHNICAL EXPLORATION OF HIGH-THROUGHPUT SATELLITES (HTS).

A high-throughput satellite is a communications system that makes use of throughput, which means the amount of data that can be sent at a time. A high-throughput satellite can be said to be a satellite that can serve efficiently within a given time and have more throughput as compared to a kind of geostationary satellite called a fixed service satellite, which provides a continuous communication link between the earth's fixed stations (ITU,2008). They are also known for broadcasting and internet connections, etc. A high throughput satellite can provide more throughput for orbital spectrum, therefore reducing cost-per-bit thus reducing the price of bandwidth. ViaSat-1 and Echo Star XVII are satellites that can provide more than one hundred gigabits of bandwidth, which is more than the amount of data produced by a fixed service satellite. ViaSat-1 had more bandwidth when it was launched than all other commercial communications satellites.

What resulted in the increase of capacity was the success of frequency reuse. Spot beam technology also enhanced frequency reuse over many smallish focused spot beams as in cellular networks, which are in a few kilometers, which define features of high-throughput satellites. Opposite to this, traditional satellite. According to Euroconsult (2017), a high-throughput satellite cannot be made with bandwidth capacity alone. While the ability to produce a large amount of bandwidth is a central feature, consumer markets are equally beneficial as a criterion. This consumer-oriented focus of high-throughput satellite has made it an advantage to the fixed service satellite, which is the traditional one that serves governments and institutions (ITU, 2017).

technology makes use of a broad single beam which can cover wide regions, places, or even entire continents.

At first, the high throughput systems satellites used earth orbit which has the same rotation as the earth at an altitude of 35,786 km called geosynchronous orbit which satellite such as satellite TV craft which shares high throughput functionality revolves in this orbit but the round-trip delay which is the delay in the time for data to be sent to the space in this case to the satellite and back to the down station. In the digital world, every second is a large, precious resource that can be used to make a significant difference in business. Satellites in this orbit, called geosynchronous orbit, provide broadcasting and internet connectivity, but due to it in built delay, make them less suitable for activities that need instant response, such as stock trades, online gaming, and video chats, including TikTok, YouTube, and Poppo, which are social media sites for streamers. Now there is a deep shift of high-throughput satellites to the lower Medium Earth orbit (MEO) and Low Earth orbit (LEO), with altitudes lower than that of the geosynchronous orbit and delays shorter than half a second. Also, satellites in the lower and medium orbits are closer, and signals don't have to travel a long distance, hence reducing power consumption and cost. More communication satellites are launched into space, which serves as coverage around the globe with higher throughput capacity (amount of data produced per second). SES O3b constellation was the first medium earth orbit (MEO) high-throughput satellite system, launched in 2013, and

by 2018, more than eighteen thousand (18,000) new low earth orbit (LEO) satellites were proposed to launch by 2025(Satellite Internet Constellation – Wikipedia, 2018).

Even though the spot beam technology resources associated are known to be costly, with that spot beam technology resource such as the cost per circuit of spot beam technology is considerably lower than that of shaped beam technology. While Ku band fixed service satellite bandwidth costs millions of dollars per gigabit per second in space, a high-throughput satellite can supply a gigabit of throughput in space for a cool price. Due to this reduced cost per bit the high-throughput satellite is always said to be of substantial advantage. High-throughput satellites, the lowest cost per bit is not always the main driver behind the design of a high-throughput satellite system; it depends on the industry it will be serving.

One of the factors why high-throughput satellite systems are designed is to provide broadband (data sent through either cable or wireless) Internet access service (point-to-point) to regions where internet access has not been reached by terrestrial technologies, where they deliver services similar to terrestrial services in terms of pricing and bandwidth. While many current high-throughput platforms were designed to serve the consumer broadband market, some are also offering services to government and enterprise markets, as well as to terrestrial cellular network operators who face growing demand for broadband backhaul to rural cell sites. For cellular backhaul, the reduced cost per bit of many high-throughput platforms creates a significantly more favorable economic model for wireless operators to use satellite for cellular voice and data backhaul. Some high-throughput satellite platforms are designed for the enterprise, telecommunication sector, and those in charge of data transfers or maritime sectors. A high-throughput satellite can furthermore support point-to-multipoint applications and even broadcast services such as DTH distribution to relatively small geographic areas served by a single spot beam. (ITU, 2017; Euroconsult, 2017).

A difference between high-throughput satellites is the fact that certain high-throughput satellites are connected to ground infrastructure through a feeder link using a regional spot beam, dictating the location of possible teleports, while other HTS satellites allow the use of any spot beam for

the location of the teleports (ITU, 2017; Euroconsult, 2017). In the latter case, the teleports can be set up in a wider area as their spotbeams' footprints cover entire continents and regions, as is the case for traditional satellites.

3.1.3 FREQUENCY REUSE

Frequency Reuse is the method of using the same radio frequency over and over throughout different networks without interference. GeeksforGeeks. (2025) Tutorialspoint. (n.d.). Frequency reuse allows networks to support more users within the same bandwidth. This is mostly done in urban areas with high demand and also reduces the need for additional spectrum, helping operators lower costs while maintaining service quality. Patsnap Eureka. (2025)

Frequency reuse improves spectral efficiency and quality of service by making the same frequencies be used over several cells at the same time. In the traditional model made for GSM systems, frequency reuse prevents interference of transmitters that share the same frequencies from interfering with one another. Within this method, the total available bandwidth is divided into multiple bands, each assigned to a specific cell. GSM systems are excluded from this method. Worldwide Interoperability for Microwave Access operators also make use of frequency reuse over different cells, making best use of the available spectrum (GeeksforGeeks, 2025; Patsnap Eureka, 2025)

3.1.4 SPOT BEAM

A spot beam, in telecommunications, is a satellite signal that is specially sent by a high-gain antenna so that it will cover only a limited area on Earth. Spot beams are used so that only Earth stations in a particular intended area can receive the signal.

For example, spot beams are on direct broadcast satellite systems such as DirecTV and Dish Network that deliver local broadcast television via satellite only to viewers in parts of North America from which those terrestrial broadcast stations originate.

Spot beams allow satellites to send different data signals using the same frequency. Because satellites have a limited number of frequencies to use, the ability to reuse a frequency for different geographical locations (without different data interfering with each other at the receiver) allows for more local channels to be carried, since the same frequency can be used in several regions.

4 COMPARATIVE ANALYSIS

4.1.1 ADVANTAGES OF SATELLITE

Satellites have a clear view of space. Satellites can collect more data, more quickly, than instruments on the ground. Satellites can see into space better than telescopes on Earth's surface. Satellites can be used for communication, such as TV and telephone signals, around the world.

Satellites offer a wider bandwidth based on SCPC or MCPC allocation types. They cover a wide area of the earth, hence the entire country or region can be covered with just one satellite. They co-exist with the terrestrial microwave line-of-sight communication system. Satellite services can be interfaced with the internet infrastructure to obtain internet service, and it is used as a GPS application.

Satellite communication is easy to install and manage the cost per VSAT site it does not incur much cost. It has a small fading margin on the order of about 3 dB, it is easy to obtain service from a single provider, and uniform service is available. It is used for voice, data, and any other information transmission.

4.1.2 DISADVANTAGES OF SATELLITES

Satellites can span about 12-15 years, so another launch has to be planned before it becomes un-operational. Satellite manufacturing requires more time, and satellite design and development require a higher cost. When a Satellite is launched, it needs to be monitored and controlled at regular periods so that it remains in orbit.

Redundant components are used in the network design, which results in more cost in the installation phase. In the case of Low Earth Orbit or Medium Earth Orbit, a large number of satellites are needed to cover the radius of Earth, and satellite visibility from the Earth is for a

very short duration, which requires a fast satellite-to-satellite handover. This makes the system very complex.

A satellite is a wide-area broadcast medium, as a transmitter antenna may be designed to offer signals to service areas as small as a city or as large as a country. Once the satellite is in position, the earth station can be installed, and communication can be established in days or hours. However, terrestrial circuits require time-consuming installations.

Advantages of Terrestrial Data Communication

Compared to satellite, terrestrial data communication systems generally have far less latency. Fiber optic or cabled systems generally have latency as low as 1ms to 20ms for local and regional links, which is important for real-time services such as video conferencing, gaming, and financial trading (Howell, 2022). They also have very much higher bandwidth capacity; newer fiber optic cables are able to have trillions of bits of information flow down them every second, allowing for extremely high capacity internet connections, 4K and 8K video streaming, and transferring massive amounts of data without loss (TelecomWorld101.com, 2024). As they are generally installed underneath the ground or run across telephone poles, they are also unaffected by rain, snow, fog or other atmospheric effects, ensuring a reliable signal in all weather conditions (National Aeronautics and Space Administration, 2023). Once installed, a terrestrial connection also has far lower operating costs per gigabyte, making unlimited plans justifiable, and many terrestrial connections have equal upload and download speeds, while satellite systems do not have any faster power-saving downloading speeds than uploading speed (Logsdon, 2026).

4.1.3 DISADVANTAGES OF TERRESTRIAL DATA COMMUNICATION

On the other hand, terrestrial networks cannot do so due to the limited scales of land offering connection access in the remote islands and deep in the woods, and also in the faraway sea, where laying down satellites is too costly for cable laying and tower mounting (Howell, 2022). Tracking network is somewhat easy as fiber optic cables and cell towers, which can be exposed to work on the ground, such as construction activities that involve excavation, damaged with animal and tree roots. It can also be ruined by earthquakes, floods, and hurricanes, and can be destroyed by terrorist attacks such as bombings. As a result of this, the whole network may not

work while satellites are not influenced by such natural events (National Aeronautics and Space Administration, 2023). Choosing to develop the terrestrial services depends on significant progress made in construction works, which requires enormous renting out of space on a tower for mounting, obtaining the license and land permit, the design process could take months or even years, and the backward rural districts wait for a long time and therefore are still covered fully (TelecomWorld101.com, 2024). Meanwhile, service providers tend to develop their network only with high subscriber levels, thus the sparsely populated territories are unserved despite the pressure demand (Logsdon, 2026).

4.1.4 CHALLENGES IN SATELLITE DATA COMMUNICATION

Satellite communications play an important role in worldwide connectivity, but it has still some intrinsic problems that affect their performance. One of the major problems faced is the atmospheric attenuation, especially for higher frequencies like Ka band. The signals are absorbed or scattered by the atmospheric elements like rain, snow, and humidity, resulting in degraded quality and interruption in the service (Panagopoulos, Arapoglou, & Cottise, 2004). This effect is called rain fade and requires some mitigation techniques, such as adaptive coding and power control for better performance.

Another crucial issue is the latency, due to satellites orbiting 36,000 kilometers above Earth in Geostationary Orbit, the signals must travel such a long distance and delay is 500-600ms. Though LEO satellites have a lower latency compared to GEO satellites, it can still be a constraint for real time applications like gaming and stock trading (Handley, 2018).

Also, the number of satellite debris orbiting Earth is increasing and is posing a serious threat of collision, which can destroy the existing satellites or disrupt any new launching of satellites. This situation can trigger the Kessler Syndrome and the increase of debris will further create a chain of collisions (Kessler & Coeur Palais, 1978).

Lastly, security and cyber security is another great issue faced by satellite communications. Satellites are susceptible to attacks like jamming, spoofing and intrusion, that can cause

interruption in communication or compromise data. Since satellites are connecting with terrestrial network and cloud systems, they require strong encryption and intrusion detection (Sharma & Gupta, 2020).

Attenuation of a wave in the atmosphere is the loss of energy carried through the air. This loss of energy in the wave is an important step in the propagation of energy in any medium. Think of this as Attenuation of a wave in the atmosphere, which is the loss of energy carried through the air. This loss of energy in the wave is an important step in the propagation of energy in any medium. Think of this as the light glinting off a stick of fog. The more it flies out, the smaller it gets as water drops scatter and absorb it. A cloud of gas, water, and other particles in the air absorbs and reflects the waves passing through it, as the light glints off a stick of fog. The more it flies out, the smaller it gets as water drops scatter and absorb it. A cloud of gas, water, and other particles in the air absorbs and reflects the waves passing through it.

5 FUTURE TRENDS

In the years to come, satellite systems will improve more than with 6G networks. This will have effect on how people get access to the internet, especially when in places which are far from the cities. 6G will have faster speed and provide us with new services. Satellites my help to bring 6G

to villages and towns where cables may not be available. As a result of this the internet reach more people.

Satellites will also improve much because of the presence of the artificial intelligence (AI). Now many satellites need people on the ground to tell them what to do. In the future, I believe satellites can make their own decisions by themselves because of the growth of Artificial intelligences. They can change their beam to send more data to a place where data is more needed. They can change frequency when there are interferences in their process. They may even find faults and try to fix small problems without waiting for the engineers. With this kind of self- work we can then call it autonomous operation.

For example, if many students in a school start a video lesson at the same time, an AI- controlled satellite can notice where there is high demand. The satellite can then focus on that particular place with a spot beam to give more bandwidth. This helps to make the lesson runs smoothly without interferences. Another example is that during a storm or disaster. Satellites with AI can quickly make changes to give emergency teams better links.

6G and satellites can be joined together with ground networks to make things better. This means that phones, base stations, and satellites will act like one system. When a person has to move from a city to a rural area, the connection can switch from a mobile tower to satellite without a big interruption. Changes in situations like this must be done smoothly. Artificial Intelligence will help manage the situation easily, so calls and video calls will not be affected

Satellites with coverage such as Low Earth Orbit satellites will play an important role in this in the future. Since Low Earth Orbit satellites are closer to the Earth compared to other forms of satellite by coverage, they give lower delay than old geostationary satellites. Many Low Earth Orbit satellites working together can give fast internet like fiber, in some cases. One factor of Low Earth Orbit is that it needs many satellites and good control. AI can be a help to manage many satellites at once. It can also plan routes, avoid collisions, and balance load between satellites.

Machine learning is a factor of AI, which will help predict where demand will be. If the system gets to know that every evening many people stream videos in a town, it can prepare by sending

more capacity there before the peak. This act of prediction saves energy and makes service better. Machine learning can also help spot strange behavior that may be a cyberattack. So that the system can block or isolate the attack fast.

Autonomous satellites can also help with the maintenance of small faults. Instead of waiting for a ground team, satellites can run some checks and report only important problems. Some satellites may even coordinate to help each other. For example, if one satellite has a failing antenna, the closest satellites can change beams to cover the area until repair.

There are also myriad ways to connect satellites using lasers. Laser links between satellites can move data very fast in space. This helps to reduce the need to send everything down to Earth first. With laser links and AI routing, data can travel more faster across space and reach the right ground station. This helps to reduce delay and makes the network more efficient and effective.

However, this future has consequences too. Making a satellite autonomous requires strong security to avoid attacks. For example, If AI systems are not safe, attackers can use them for wrong things. False commands can cause big trouble. So, security must be part of the design. Also, a lot of LEO satellites means more objects in space. We should manage space traffic and debris carefully. AI can help predict collisions free plan safe moves, but rules and cooperation between countries are needed.

Another issue is the raising of price. developing and moving many satellites into the sky is costly. Countries and companies invest a lot. For poorer areas or regions, cost may slow adoption. But some shared systems and partnerships can help reduce cost. For example, regional hubs can buy their services from satellite providers instead of each country building its own system.

For students and young people, the change can be good. Better internet means more access to online classes, research, and global information. Farmers can use smart sensors to check soil and weather. Doctors can consult specialists far away. Small businesses can sell to customers in other countries. These benefits depend on good planning and fair access.

In short, the future will see satellites and 6G working together with AI to make networks faster and smarter. Satellites will not only carry signals; they will think and act to keep the network

running. This will bring big benefits, but also new risks and costs. To get the best results, engineers, governments, and communities must work together. They must build secure AI systems, manage space safely, and make sure services reach those who need them most.

6 REFERENCES

- Allis, C. D., et al. (2021). Survey on 6G frontiers: Trends, applications, requirements, technologies and future research. *IEEE Open Journal of the Communications Society*, 2, 836–886.
- Antesky. (2021, May 14). Application and development of Ka-band antenna. <https://antesky.com/application-and-development-of-ka-band/>
- AudioDramaProduction. (2026, January 28). FM (Frequency Modulation). <https://audiodramaproduction.com/radio-broadcasting-glossary/fm-frequency-modulation/>
- Bhattacharjee, D., Iqbal, W., Singla, A., & Singla, P. (2019). Network topology design at 27,000 km/hour. *Proceedings of the ACM CoNEXT Conference*.
- Britannica. (1998). Modulation. *Encyclopaedia Britannica*. <https://www.britannica.com/technology/modulation-communications>
- Britannica. (2025). Low Earth orbit. *Encyclopædia Britannica*. <https://www.britannica.com/technology/low-Earth-orbit>
- Britannica Editors. (2024). Satellite communication: Development of satellite communication. In *Encyclopedia Britannica*. <https://www.britannica.com/technology/satellite-communication/Development-of-satellite-communication>
- Cadence PCB. (2023). Ka-band radar frequency applications. <https://resources.pcb.cadence.com/blog/2023-ka-band-radar-frequency-applications>
- Calcutta, D., & Tetley, L. (1994). *Satellite communications: Principles and applications*. Elsevier.
- Calderon, M. (2014). *Handbook of satellite orbits: From Kepler to GPS*. Springer.
- Chobotov, V. A. (2002). *Orbital mechanics* (3rd ed.). American Institute of Aeronautics and Astronautics.
- Clarke, A. C. (1945). Extra-terrestrial relays: Can rocket stations give world-wide radio coverage? *Wireless World*, 51(10), 305–308.
- Curtis, H. D. (2013). *Orbital mechanics for engineering students* (3rd ed.). Butterworth-Heinemann.
- Dang, S., Amin, O., Shihada, B., & Alouini, M. S. (2020). What should 6G be? *Nature Electronics*, 3(1), 20–29. <https://doi.org/10.1038/s41928-019-0355-6>

del Portillo, I., Cameron, B. G., & Crawley, E. F. (2019). A technical comparison of three LEO satellite network constellations. *Acta Astronautica*, 159, 123–135.

Diffen. (n.d.). AM vs FM. https://www.diffen.com/difference/AM_vs_FM

Dummies. (2025). AM broadcast frequency spectrum. <https://www.dummies.com/article/business-careers-money/careers/trades-tech-engineering-careers/am-broadcast-frequency-spectrum-165549/>

Elbert, B. R. (2002). *The satellite communication applications handbook*. Artech House.

Elbert, B. R. (2004). *Introduction to satellite communication* (3rd ed.). Artech House.

Encyclopedia.com. (n.d.). Satellite communication. <https://www.encyclopedia.com>

eoPortal. (2023). INCUS (INvestigation of Convective UpdraftS). <https://www.eoportal.org/satellite-missions/incus>

European Space Agency. (2022). ESA's annual space environment report. ESA Space Debris Office.

European Space Agency. (2023). Space debris by the numbers. ESA Space Debris Office.

Federal Communications Commission. (2015, December 11). Why AM stations must reduce power, change operations, or cease broadcasting at night. <https://www.fcc.gov/media/radio/am-stations-at-night>

Federal Communications Commission. (2020). FCC authorizes Amazon's Project Kuiper satellite constellation. FCC.

Freeman, R. L. (2007). *Radio system design for telecommunications* (3rd ed.). Wiley-Interscience.

Garcia, M. (2021). About the space station: Facts and figures. NASA Human Spaceflight.

GeeksforGeeks. (2025, July 12). Frequency reuse. <https://www.geeksforgeeks.org/computer-networks/frequency-reuse/>

Giordani, M., Polese, M., Mezzavilla, M., Rangan, S., & Zorzi, M. (2020). Toward 6G networks: Use cases and technologies. *IEEE Communications Magazine*, 58(3), 55–61.

Gui, G., Liu, M., Tang, F., Kato, N., & Adachi, F. (2020). 6G: Opening new horizons for integration of comfort, security, and intelligence. *IEEE Wireless Communications*, 27(5), 126–132.

Handley, M. (2018). Delay is not an option: Low latency routing in space. *Proceedings of the 17th ACM Workshop on Hot Topics in Networks (HotNets)*.

Hastings, D., & Garrett, H. (1996). *Spacecraft-environment interactions*. Cambridge University Press.

Haykin, S. (2001). *Communication systems* (4th ed.). John Wiley & Sons.

History of Information. (n.d.). Communication by geosynchronous satellites predicted (October 1945).
<https://wayback.archive-it.org/all/20120703005613/http://www.historyofinformation.com/expanded.php?id=2509>

Howell, E. (2022, January 17). What is a satellite? Space.com.
<https://www.space.com/24839-satellites.html>

IB-Lenhardt. (n.d.). Signal modulation – types and applications.
<https://ib-lenhardt.com/kb/glossary/signal-modulation>

IEEE Communications Society. (2022). First practical concept of satellite communications.
<https://www.comsoc.org/node/19071>

Intergovernmental Panel on Climate Change. (2021). *Climate change 2021: The physical science basis*. Cambridge University Press.

International Telecommunication Union. (1997). Ka band requirements for geostationary satellite systems. <https://www.itu.int/newsarchive/press/WRC97/Ka-band.html>

International Telecommunication Union. (2008). Arthur C. Clarke: A visionary of the space age. ITU News.
<https://www.itu.int/itunews/manager/display.asp?lang=en&year=2008&issue=03&ipage=Arthur-Clarke&ext=html>

International Telecommunication Union. (2020). *Handbook on satellite communications* (4th ed.). ITU.

Jacchia, L. G. (1971). Revised static models of the thermosphere and exosphere with empirical temperature profiles. Smithsonian Astrophysical Observatory.

Japan Aerospace Exploration Agency. (2018). Super low altitude test satellite "Tsubame" (SLATS): World's lowest altitude observation. JAXA.

Kaur, J., Khan, M. A., Iftikhar, M., Imran, M., & Ul Haqq, Q. E. (2021). Machine learning techniques for 5G and beyond. *IEEE Access*, 9, 23472–23488.
<https://doi.org/10.1109/ACCESS.2021.3051557>

Kessler, D. J., & Cour-Palais, B. G. (1978). Collision frequency of artificial satellites: The creation of a debris belt. *Journal of Geophysical Research*, 83(A6), 2637–2646.

King-Hele, D. (1987). *Satellite orbits in an atmosphere: Theory and applications*. Blackie and Son.

Kolawole, M. O. (2013). *Satellite communication engineering* (2nd ed.). CRC Press.

Larson, W. J., & Pranke, L. K. (1999). *Human spaceflight: Mission analysis and design*. McGraw-Hill.

Letaief, K. B., Chen, W., Shi, Y., Zhang, J., & Zhang, Y. J. A. (2019). The roadmap to 6G: AI empowered wireless networks. *IEEE Communications Magazine*, 57(8), 84–90.

Liu, Y., Qin, Z., Elakashlan, M., Gao, Y., & Li, G. Y. (2021). Satellite-terrestrial integrated networks for 6G: Architecture, challenges, and opportunities. *IEEE Internet of Things Journal*, 8(18), 13176–13192.

Lutz, E., Werner, M., & Jahn, A. (2000). *Satellite systems for personal and broadband communications*. Springer.

Maral, G., & Bousquet, M. (2020). *Satellite communications systems: Systems, techniques and technology* (6th ed.). Wiley.

NASA. (1964). Syncom 3. NASA Space Science Data Coordinated Archive. <https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1964-047A>

NASA. (2022). International Space Station facts and figures. NASA.

NASA. (2023). Earth's upper atmosphere. NASA Science.

National Aeronautics and Space Administration. (2023, September 26). What is a satellite? NASA Space Place. <https://spaceplace.nasa.gov/satellite/en/>

Noman, H. M. F., et al. (2023). Machine learning empowered emerging wireless networks in 6G: Recent advancements, challenges & future trends. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2023.XXXXXXX>

Panagopoulos, A. D., Arapoglou, P. D., & Cottis, P. G. (2004). Satellite communications at Ku, Ka, and V bands: Propagation impairments and mitigation techniques. *IEEE Communications Surveys & Tutorials*, 6(3), 2–14.

Pelton, J. N. (2019). The growth and expansion of precise navigation and timing. In *Space 2.0*. Springer. https://doi.org/10.1007/978-3-030-15281-9_4

Pelton, J. N., Madry, S., & Camacho-Lara, S. (Eds.). (2017). Handbook of satellite applications (2nd ed.). Springer. <https://link.springer.com/referencework/10.1007/978-3-319-23386-4>

Perez, R. (1998). Wireless communications design handbook: Volume I – Space. Academic Press.

Pratt, T., Bostian, C. W., & Allnutt, J. E. (2003). Satellite communications (2nd ed.). Wiley.

Rees, W. G. (2013). Physical principles of remote sensing (3rd ed.). Cambridge University Press.

Roddy, D. (2019). Satellite communications (5th ed.). McGraw Hill.

Saeed, N., et al. (2021). Point-to-point communication in integrated satellite-aerial 6G networks: State-of-the-art and future challenges. IEEE Open Journal of the Communications Society, 2, 1505–1525. <https://doi.org/10.1109/OJCOMS.2021.3071496>

Sellers, J. J. (2005). Understanding space: An introduction to astronautics (3rd ed.). McGraw-Hill.

SES. (2025). What is medium Earth orbit (MEO)? SES. <https://www.ses.com/what-is-meo>

Sharma, R., & Gupta, R. (2020). Cybersecurity challenges in satellite communication networks. International Journal of Satellite Communications and Networking, 38(5), 437–450.

TelecomWorld101.com. (2024, February 27). The role of satellites in global connectivity. <https://telecomworld101.com/satellites-global-connectivity/>

Trimble, V., & Aschwanden, M. J. (2001). Astrophysics in 2000. Publications of the Astronomical Society of the Pacific, 113(787), 1025–1114. <https://doi.org/10.1086/322844>

Tutorialspoint. (n.d.). Principles of communication – modulation. https://www.tutorialspoint.com/principles_of_communication/principles_of_communication_modulation.htm

U.S. Code of Federal Regulations. (n.d.). 47 CFR § 73.14 – AM broadcast definitions. Legal Information Institute, Cornell Law School. <https://www.law.cornell.edu/cfr/text/47/73.14>

University of Waikato. (2022, August 10). History of satellites – timeline. Science Learning Hub. <https://www.sciencelearn.org.nz/resources/1905-history-of-satellites-timeline>

Vallado, D. A., & Finkleman, D. (2014). A critical assessment of satellite drag and atmospheric density modeling. Acta Astronautica, 95, 141–165.

Varrall, G. (2023). 5G and satellite RF and optical integration. Artech House.

Wertz, J. R., Everett, D. F., & Puschell, J. J. (2011). Space mission engineering: The new SMAD. Microcosm Press.

Wikipedia. (n.d.). FM broadcasting. Wikimedia Foundation.

https://en.wikipedia.org/wiki/FM_broadcasting

Wikipedia. (n.d.). Signal modulation. Wikimedia Foundation.

https://en.wikipedia.org/wiki/Signal_modulation

Xu, X., Zhang, J., & Wang, J. (2022). Artificial intelligence empowered satellite communication networks: Toward autonomous operations. *IEEE Wireless Communications*, 29(1), 144–151.

Zhang, C., & Dai, L. (2023). Enhancing 6G-satellite network integration with artificial intelligence: A future communication paradigm. In *Proceedings of the IEEE International Conference on Communications (ICC)* (pp. 1–6).

Zhang, Z., Xiao, Y., Ma, Z., Xiao, M., Ding, Z., Lei, X., ... & Poor, H. V. (2019). 6G wireless networks: Vision, requirements, architecture, and key technologies. *IEEE Vehicular Technology Magazine*, 14(3), 28–41.4.